A Physics-inspired Spiking Neural Network For Explainable Decision-support in Power Grid Operation

Zuoxu Wu†
School of Engineering
Westlake University
Hangzhou, China
wuzuoxu@westlake.edu.cn

Wenxuan Pan†
School of Engineering
Westlake University
Hangzhou, China
panwenxuan@westlake.edu.cn

Yaochu Jin*
School of Engineering
Westlake University
Hangzhou, China
jinyaochu@westlake.edu.cn

Yanxu Dong
State Grid Fujian Information & Telecommunication Company
Fuzhou, China
515663554@qq.com

Mengjie Shi
China Electric Power Research Institute
Fuzhou, China
shimengjie273@163.com

Cuncun Shi
China Electric Power Research Institute
Fuzhou, China
ccshi1012@163.com

***Abstract*—The rapid integration of renewable energy introduces unprecedented stochasticity to modern power grids, necessitating a paradigm shift from deterministic state prediction to reliable operational risk assessment. In this paper, we propose a novel Neuro-symbolic Closed-loop Learning Framework designed to evaluate a unified Grid Operational Risk Score (GORS). Instead of relying on purely data-driven black-box models, we employ a compact single-compartment Leaky Integrate-and-Fire (LIF) Spiking Neural Network (SNN) to capture temporal grid dynamics. To encourage operational consistency and logical reliability of the risk assessment, we introduce a differentiable neuro-symbolic rule layer that translates discrete expert heuristics into continuous boundary targets, alongside a physics-constrained learning layer that penalizes operational consistency violations. Furthermore, we develop an unsupervised closed-loop refinement mechanism that leverages internal consistency residuals—comprising symbolic and physical violations—to dynamically inject corrective feedback currents, reducing dependence on manually annotated risk labels. Experiments on the multi-region PSML dataset show that the proposed framework provides interpretable and structurally consistent risk assessment under renewable and weather-induced uncertainty, highlighting a trade-off between predictive fitting accuracy and operational reliability.**

Keywords-component: Spiking Neural Networks; Neuro-symbolic Artificial Intelligence; Smart Grid Decision Support

I. INTRODUCTION

The rapid integration of renewable energy resources, distributed generation, and flexible loads is fundamentally transforming modern power systems [26,27]. Compared with conventional grids, future power systems are characterized by stronger uncertainty, higher operational variability, and increasingly complex interactions among weather conditions, renewable generation, energy storage, and electricity demand [26,27,29.] As a result, grid operators are required to continuously interpret large volumes of heterogeneous temporal observations and respond to rapidly evolving operating conditions in real time [26,29]. In practical scenarios, system operators must evaluate the current operating state, incorporate domain knowledge, and determine appropriate intentions in uncertain environments. Consequently, the central challenge is no longer merely how to predict future observations but how to transform multi-modal operational observations into reliable, interpretable, and operationally consistent decision recommendations.

Despite the widespread application of data-driven intelligence methodologies—such as Long Short-Term Memory (LSTM) networks and Transformers [31,35]—these models face critical bottlenecks in real-world grid dispatch. First, they are frequently criticized as "black boxes" owing to their poor interpretability. Second, and more importantly, pure data-driven models lack the explicit integration of fundamental physical laws [19,20,21] (e.g., power balance) and heuristic operational logic (e.g., weather-induced stress). Consequently, when confronted with out-of-distribution extreme weather anomalies, these models are prone to generating physically infeasible or logically absurd assessments, undermining operator trust.

To capture the high-frequency transient dynamics of the grid, Spiking Neural Networks (SNNs) have emerged as a promising connectionist paradigm [1,5,10]. Communicating via discrete spike trains, SNNs inherently excel at processing spatiotemporal dynamics with exceptional efficiency [5, 6, 10]. Simultaneously, Neuro-symbolic learning presents a pioneering frontier that unifies the perception strengths of neural networks with the rigorous reasoning capabilities of symbolic logic [11,13,14,17]. However, integrating neuro-symbolic reasoning

† These authors contributed equally to this work. * Corresponding author.

with spiking temporal representation learning for intelligent power grid operation remains largely unexplored [11,14,17].

To bridge these gaps, we propose a novel Neuro-symbolic Closed-loop Learning Framework designed to evaluate a unified Grid Operational Risk Score (GORS). Rather than attempting to output deterministic control commands—which are often unavailable as ground-truth labels in SCADA datasets—our framework estimates a continuous operational risk score. The architecture encodes multi-modal data into spike trains, utilizes a single-compartment Leaky Integrate-and-Fire (LIF) SNN [1,3,4] to extract dynamic risk features, and rigorously validates these features through a differentiable neuro-symbolic rule layer and a physics-constrained learning layer. Uniquely, the framework leverages the consistency residuals from symbolic and physical violations to dynamically inject corrective feedback currents, achieving self-guided state refinement.

The main contributions of this paper are summarized as follows:

1. A spatiotemporal spiking neural network (SNN) backbone is proposed to capture temporal grid dynamics. Through rigorous empirical architecture search, a single-compartment LIF model is selected to efficiently disentangle heterogeneous meteorological and electrical perturbations without introducing unwarranted optimization instability.
2. A knowledge-guided neuro-symbolic and physics-inspired learning strategy is established. By translating discrete expert heuristics into continuous boundary targets and penalizing topology-free operational consistency violations, the framework encourages the estimated risks to comply with both operational logic and available physical consistency information..
3. An unsupervised closed-loop refinement mechanism is designed to reduce symbolic and physical inconsistencies. The mechanism transforms internal consistency residuals into feedback currents, thereby guiding the spiking dynamics toward more reliable risk representations.

While dense architectures may achieve marginally lower regression error on this task, the proposed framework uniquely provides verifiable physical compliance, symbolic interpretability, and closed-loop consistency—properties that are critically absent in conventional black-box predictors.

II. *Related Work*

In this section, we provide a comprehensive review of the foundational literature and recent advancements that underpin our proposed framework. We examine the evolution of data-driven power grid intelligence, explore the emerging fusion of brain-inspired computing and neuro-symbolic logic, and analyze the state-of-the-art in physics-informed learning paradigms.

A. Data-driven Power Grid Intelligence

The modern power grid is undergoing a profound transition with the integration of renewable energy, heightening operational stochasticity. Traditional model-based approaches are increasingly supplemented by data-driven methodologies, ranging from statistical methods (e.g., ARIMA) to deep learning architectures (e.g., LSTM, Transformers, and GNNs) for tasks like load forecasting and stability assessment [27, 31]. However, these approaches primarily formulate grid intelligence as a direct input-output prediction problem. They are frequently criticized as "black boxes" and rarely incorporate explicit operational knowledge or fundamental physical boundaries into the decision process.

B. Brain-inspired and Knowledge-guided Learning

To capture high-frequency transient dynamics, Spiking Neural Networks (SNNs) offer exceptional energy efficiency and intrinsic temporal dynamics via discrete spike trains [1, 5]. While deployed in neuromorphic vision, their application in smart grids remains largely restricted to prediction paradigms without autonomous decision support. Concurrently, neuro-symbolic learning unifies connectionist perception with formal symbolic logic, enabling gradient-based optimization of discrete rules [13, 17]. Furthermore, physics-informed AI introduces critical physical boundaries into neural optimization [20, 25]. Despite their potential, existing methods fail to concurrently reconcile spiking temporal dynamics, continuous physical equations, and discrete operational heuristics. Bridging this gap is the primary objective of our proposed framework.

III. *Proposed Framework*

Our goal is to learn a mapping function $f_\theta: X_{t-T} \rightarrow y_t$, where X_{t-T} denotes the historical multimodal grid observations, including load, renewable generation, and weather variables, and $y_t \in [0,1]$ represents the unified Grid Operational Risk Score (GORS). The proposed framework consists of three coupled components: a spatiotemporal spiking neural encoder, a knowledge-guided reasoning module, and a closed-loop refinement mechanism. The final GORS is not intended to directly replace dispatch decisions, but to provide an interpretable operational awareness indicator for grid operators.

A. Overall Architecture

The intelligent decision-support problem is formulated as a spatiotemporal risk assessment task over a continuous observation window. Let the multimodal grid state at time step (t) be defined as X_t:

$$X_t = [L_t, R_t, W_t]$$

where L_t denotes load dynamics, R_t denotes renewable generation features such as wind and solar generation, and W_t denotes meteorological variables such as temperature, wind speed, humidity, and irradiance. Given a historical temporal window $X_{t-T:t}$, the overarching goal is to learn a mapping function f_θ parameterized by SNN weights θ:

$$f_\theta: X_{t-T:t} \rightarrow y_t$$

where$y_t \in [0,1]$ represents the Unified Grid Operational Risk Score (GORS). A higher score implies severe grid stress necessitating immediate operator intervention. To ensure the reliability of y_t, the mapping must be jointly optimized to satisfy historical data patterns, heuristic expert rules, and physical boundaries $\mathcal{P}$.

B. Dynamic Representation Learning

To capture temporal fluctuations in multimodal grid observations, the continuous input sequence is first converted into sparse spike trains through stochastic rate coding. For the i-th normalized feature $x_i(t)$, the binary spike event $s_i(t) \in [0,1]$ is generated as

$$s_i(t) = \begin{cases} 1, if \quad rand() < \phi(x_i(t)) \\ 0, \qquad\qquad otherwise \end{cases} \tag{1}$$

where $\phi(\cdot)$ maps the normalized input to a firing probability. This encoding transforms load, renewable, and weather variations into event-driven temporal signals.The encoded spike trains are then processed by a single-compartment Leaky Integrate-and-Fire (LIF) spiking neural encoder. Compared with more complex multi-compartment neuron models, the single-compartment LIF structure provides a stable and efficient backbone for this regression-oriented risk assessment task. The input current is computed from the encoded spike stream as $I_t = W_s s(t)$,and the membrane potential evolves according to

$$v_t = \lambda v_{t-1}(1 - h_{t-1}) + I_t \tag{2}$$

where $(\lambda \in (0,1))$ is the membrane decay factor. A spike is emitted when the membrane potential exceeds the threshold:

$$h_t = H(v_t - v_{\text{th}}) \tag{3}$$

The temporal aggregation of hidden spike activities forms the dynamic representation z_t, which is then passed to the knowledge-guided reasoning module.

C. Knowledge-guided Closed-loop Learning

The knowledge-guided module integrates symbolic rules, physical consistency constraints, and residual-driven feedback into a unified closed-loop process. First, the neuro-symbolic rule layer converts operational heuristics into differentiable soft constraints. For the k-th rule, an expert target $e_k(t)$ is automatically generated from the input state using a monotonic mapping:

$$e_k(t) = \sigma(\alpha_k q_k(X_t)) \tag{4}$$

where $q_k(\cdot)$ denotes the rule-specific risk indicator, such as temperature stress, load ramp, renewable fluctuation, or net-load variation. Let $a_k(t)$ be the corresponding risk feature predicted from z_t. The rule gap is defined as

$$g_k(t) = |a_k(t) - e_k(t)| \tag{5}$$

D. Optimization Objective

The entire framework is trained end-to-end using a unified objective that balances predictive accuracy, symbolic consistency, physical feasibility, and feedback regularization:

The gap is transformed into a differentiable soft truth value:

$$T_k(t) = \sigma(-\beta g_k(t)) \tag{6}$$

where β controls the sharpness of the truth assignment. The overall symbolic satisfaction is aggregated through a product t-norm,

$$T(t) = \prod_{k=1}^{K} T_k(t) \tag{7}$$

and the symbolic loss is defined as

$$L_{\text{sym}} = -\log(T(t) + \epsilon) \tag{8}$$

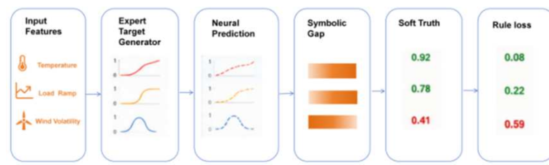

Figure 1. Neuro-symbolic Rule Layer

In parallel, the physics-constrained layer penalizes operational inconsistency using three topology-free constraints available from the PSML variables: power balance consistency, ramp-rate consistency, and renewable capacity consistency. The total physical penalty is written as:

$$L_{\text{phy}} = L_{\text{pb}} + L_{\text{ramp}} + L_{\text{ren}}. \tag{9}$$

Since the PSML dataset does not contain full network topology or AC power-flow states, these terms are implemented as operational consistency regularizers rather than full power-flow constraints.

As illustrated in Fig. 2, the physics-constrained closed-loop refinement module converts symbolic and physical violations into an internal feedback signal. Specifically, the symbolic loss and physical consistency loss are first aggregated into a residual term:

$$r_t = L_{\text{sym}} + L_{\text{phy}} \tag{10}$$

The residual is projected into a feedback current through a learnable alignment matrix:

$$I_{\text{fb}} = W_f r_t \tag{11}$$

This feedback current is injected into the next LIF update,

$$I_{t+1} \leftarrow I_{t+1} + I_{\text{fb}} \tag{12}$$

so that the spiking dynamics are iteratively guided toward a representation that is more consistent with both operational rules and physical constraints.

$$L_{\text{total}} = L_{\text{data}} + \lambda_1 L_{\text{sym}} + \lambda_2 L_{\text{phy}} + \lambda_3 L_{\text{fb}} \tag{13}$$

where L_{data} denotes the mean squared error between the predicted GORS and the pseudo-risk target, L_{sym} denotes the symbolic rule loss, L_{phy} denotes the physical consistency penalty, and L_{fb} regularizes the magnitude of the feedback current. The hyperparameters $\lambda_1, \lambda_2, and\ \lambda_3$control the relative contributions of the three knowledge-guided terms.

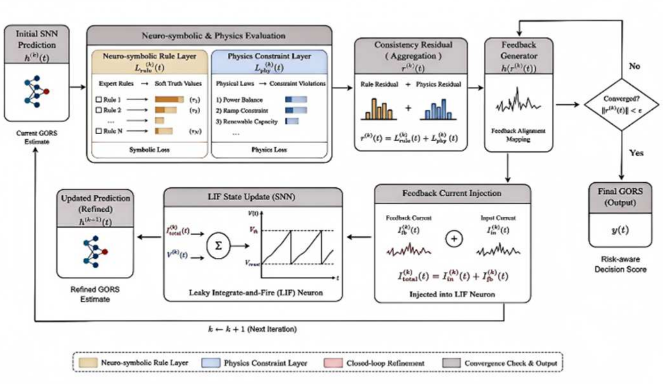

Figure 2. Physics-constrained Closed-loop Refinement

IV. *EXPERIMENTS*

In this section, we evaluate the proposed framework on the PSML dataset from three perspectives. First, we compare its predictive behavior with representative temporal models. Second, we examine whether the symbolic, physical, and feedback components contribute to structural consistency. Third, we provide scenario-level analysis to illustrate how the predicted GORS supports interpretable grid operation assessment.

A. Experimental Setup

We validate the proposed framework on the minute-level regional time-series files provided in the PSML dataset [38], which contains minute-resolution multi-region time-series data for decarbonized power grids. The dataset includes 11 heterogeneous variables, covering electrical features such as load, wind generation, and solar generation, as well as meteorological features such as wind speed, temperature, humidity, irradiance, and solar zenith angle.

To reduce spatiotemporal leakage, the dataset is divided into geographically disjoint training, validation, and test subsets. The training set contains 26 zones, the validation set contains 10 zones, and the test set contains 30 zones. All variables are standardized using z-score normalization. A sliding window with length (T=96) minutes and stride 96 is used to construct temporal samples. Since explicit operational risk labels are unavailable in PSML, we construct a pseudo Grid Operational Risk Score (GORS) as a weak supervision signal. Five operational variables are selected from the 11-dimensional PSML features, including load power, wind generation, solar generation, wind speed, and temperature. The input variables are first standardized using per-zone z-score normalization, and the stress indicators are calculated from the relative variations between consecutive 96-minute windows:

$$y_t = \sigma\left(2\left(0.5\frac{|\triangle L_t^{net}|}{\sigma_L} + 0.3\frac{|\triangle R_t|}{\sigma_R} + 0.12\frac{|W_t-\mu_W|}{\sigma_W} + 0.08\frac{|T_t-\mu_T|}{\sigma_T} - 1\right)\right) \quad (14)$$

where $L^{net}, R, W,$ and T denote net load, renewable generation, wind speed, and temperature, respectively. The weights are assigned according to operational relevance, with higher importance given to electrical imbalance and renewable uncertainty. The sigmoid transformation maps the aggregated stress indicator into [0,1]. GORS is used as a reproducible pseudo-risk target rather than a ground-truth operational risk label. The weights are chosen according to their operational relevance: net-load variation directly reflects the balance pressure between demand and renewable supply, renewable fluctuation captures uncertainty induced by wind and solar generation, while wind-speed and temperature terms represent weather-induced stress factors. The final score is clipped to [0,1], where a larger value indicates a more stressed operating condition. We compare the proposed framework with four representative baselines: LSTM, Transformer, TCN, and a vanilla SNN-LIF model. All baselines are trained with the same input features and optimized using mean squared error loss. The proposed model additionally incorporates symbolic rule loss, physics consistency loss, and feedback regularization. All models are implemented in PyTorch and trained using the Adam optimizer with a learning rate of (1×10^{-3}), batch size 32, and a maximum of 50 epochs.The evaluation metrics include RMSE, MAE, and Spearman's rank correlation coefficient (ρ) for risk prediction. In addition, we report symbolic rule trust and physics violation score to evaluate whether the predicted risk satisfies operational logic and physical consistency.

B. Performance Evaluation

Table I reports the test-set performance of all evaluated methods on the GORS prediction task. Dense temporal models such as LSTM, Transformer, and TCN achieve strong regression accuracy, reflecting their capability in fitting temporal patterns from historical data. The SNN-LIF baseline also shows competitive temporal modeling performance with a smaller parameter footprint, indicating that event-driven spiking computation can capture useful grid dynamics.The proposed GORS framework is designed not only for regression accuracy but also for structurally consistent risk assessment. Therefore, its value should be interpreted from both predictive and operational perspectives. While purely data-driven models may achieve lower RMSE under the pseudo-risk target, they do not explicitly enforce expert rules or physical consistency. In contrast, the proposed framework introduces additional constraints that improve rule compliance and reduce physically inconsistent risk assessments.

TABLE I SUMMARIZES THE TEST-SET PERFORMANCE OF ALL EVALUATED METHODOLOGIES ON THE GORS PREDICTION TASK.

Method	RMSE ↓	MAE ↓	Spearman ρ ↑
LSTM	0.060	0.184	0.507
Transformer	0.044	0.179	0.533
TCN	0.062	0.183	0.512
SNN-LIF	0.071	0.176	0.524
GORS (Ours)	0.353	0.302	0.080

Table II presents the ablation study. Removing the neuro-symbolic rule layer weakens rule consistency, indicating that differentiable expert rules provide useful operational guidance. Removing the physics-constrained layer increases the physical violation score, showing that the physical regularizers help suppress unrealistic risk estimation. Removing the closed-loop feedback module degrades the stability of the final prediction, suggesting that residual-driven feedback contributes to more consistent representations.

TABLE II ABLATION STUDY OF THE PROPOSED GORS FRAMEWORK

Variant	RMSE ↓	ρ ↑	Rule Trust T	Phys. Violations
GORS (Full)	0.3529	0.302	0.302	0.080
w/o Symbolic Rules	0.3538	0.291	—	0.079
w/o Physics Constraints	0.2903	0.260	0.260	0.140
w/o Closed-loop Feedback	0.3602	0.302	0.302	0.076

It is worth noting that the full model does not always obtain the lowest RMSE among all ablation variants. For example, removing the physics-inspired consistency constraints leads to a lower RMSE, because the model is less restricted and can more freely fit the pseudo-risk target. However, this improvement in curve fitting is accompanied by a higher physical violation score, indicating weaker operational consistency. This result reveals an accuracy–consistency trade-off: purely minimizing regression error may improve numerical fitting to the pseudo target, but it may also produce risk estimates that violate available operational constraints. Therefore, the proposed framework should be evaluated not only by RMSE and MAE, but also by rule trust and physical violation metrics.

The proposed framework adopts a single-compartment LIF neuron as the spiking backbone. Compared with more complex multi-compartment neuron models, the single-compartment structure provides a better balance between temporal representation capability and optimization stability for the regression-oriented GORS estimation task.

C. Interpretability and Operational Scenario Analysis

Beyond numerical prediction, an important advantage of the proposed framework is that the predicted GORS can be interpreted through symbolic rule truth values and physical violation signals. Fig. 3 illustrates a representative 24-hour operating window. The input panel shows load, renewable generation, and meteorological changes; the GORS panel shows the corresponding risk trajectory; the rule-truth panel identifies which symbolic rules are activated; and the spike-activity panel visualizes the response of the SNN encoder.

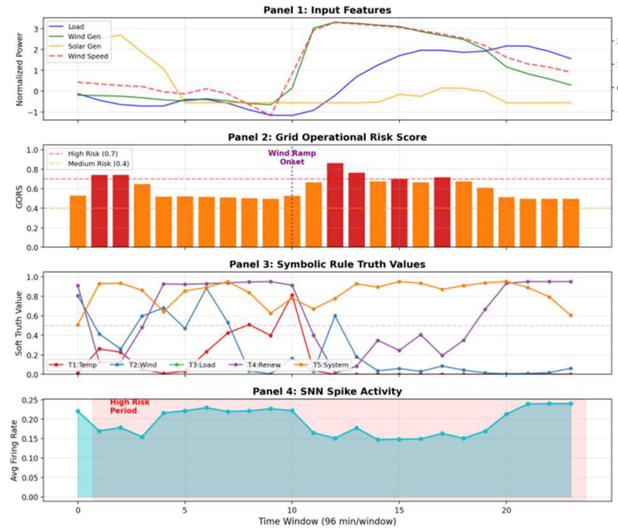

Figure 3. 24-hour case study

During stable operating periods, the predicted GORS remains low and most symbolic rules maintain high truth values. When rapid renewable fluctuation or weather-induced disturbance occurs, the corresponding rule truth decreases and the GORS increases accordingly. This behavior indicates that the model does not merely output a scalar risk value, but also provides interpretable evidence about why a specific period is considered risky.

Beyond single-case visualization, we further conduct statistical scenario analysis over multiple operating periods in the test set. The test samples are categorized into normal operation, renewable ramp events, and extreme weather periods according to renewable fluctuation and meteorological indicators. The results in Table III show that GORS exhibits consistent increasing trends under higher operational stress, while symbolic rule trust decreases and consistency penalties increase, indicating that the proposed framework captures meaningful operational variations.

TABLE III STATISTICAL ANALYSIS OF GORS

Condition	Windows	Mean GORS	Std	Rule Trust
Normal operation	488	0.524	0.035	0.566
Renewable ramp events	434	0.616	0.108	0.415
Extreme weather periods	15	0.759	0.144	0.425

V. CONCLUSIONS

In this paper, we proposed a neuro-symbolic closed-loop learning framework for power grid decision support. By combining spatiotemporal SNN representation, neuro-symbolic constraints, physics-inspired consistency regularization, and residual feedback, the framework provides interpretable GORS estimation under renewable uncertainty. Experiments on PSML demonstrate improved operational interpretability and consistency, suggesting the potential of knowledge-guided spiking models for reliable grid intelligence.

ACKNOWLEDGMENT

This work was supported by the Science and Technology Project of State Grid Corporation of China under Grant No. 521300250036-335-WLCY.

REFERENCES

[1] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Networks*, vol. 10, no. 9, pp. 1659-1671, 1997.

[2] E. M. Izhikevich, "Simple model of spiking neurons," *IEEE Transactions on Neural Networks*, vol. 14, no. 6, pp. 1569-1572, 2003.

[3] Y. Wu, L. Deng, G. Li, J. Zhu, and L. Shi, "Spatio-temporal backpropagation for training high-performance spiking neural networks," *Frontiers in Neuroscience*, vol. 12, p. 331, 2018.

[4] E. O. Neftci, H. Mostafa, and F. Zenke, "Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 51-63, 2019.

[5] K. Roy, A. Jaiswal, and P. Panda, "Towards spike-based machine intelligence with neuromorphic computing," *Nature*, vol. 575, no. 7784, pp. 607-617, 2019.

[6] A. Tavanaei, M. Ghodrati, S. R. Kheradpisheh, T. Masquelier, and A. Maida, "Deep learning in spiking neural networks," *Neural Networks*, vol. 111, pp. 47-63, 2019.

[7] Z. Zhou, Y. Zhu, C. He, Y. Wang, S. Yan, J. Tian, and L. Yuan, "Spikformer: When spiking neural network meets transformer," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.

[8] W. Fang, Z. Yu, Y. Chen, T. Masquelier, T. Huang, and Y. Tian, "Incorporating learnable membrane time constant to enhance learning of spiking neural networks," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[9] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers in Computational Neuroscience*, vol. 9, p. 99, 2015.

[10] J. K. Eshraghian, M. Ward, E. Neftci, et al., "Training spiking neural networks using lessons from deep learning," *Proceedings of the IEEE*, vol. 111, no. 9, pp. 1016-1054, 2023.

[11] A. d'Avila Garcez, M. Gori, L. C. Lamb, L. Serafini, M. Spranger, and S. N. Tran, "Neural-symbolic computing: An effective methodology for principled AI," *arXiv preprint arXiv:1905.06088*, 2019.

[12] J. Xu, Z. Zhang, T. Friedman, Y. Liang, and G. Van den Broeck, "A semantic loss function for deep learning with symbolic knowledge," in Proceedings of the 35th International Conference on Machine Learning (ICML), Proceedings of Machine Learning Research, vol. 80, pp. 5502–5511, 2018.

[13] R. Manhaeve, A. Dumančić, A. Kimmig, T. Demeester, and L. De Raedt, "DeepProbLog: Neural probabilistic logic programming," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[14] L. De Raedt, A. Dumančić, R. Manhaeve, and G. Marra, "From statistical relational to neuro-symbolic artificial intelligence," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.

[15] Z. Hu, X. Ma, Z. Liu, E. Hovy, and E. Xing, "Harnessing deep neural networks with logic rules," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2016.

[16] T. Rocktäschel and S. Riedel, "End-to-end differentiable proving," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[17] S. Badreddine, A. d'Avila Garcez, L. Serafini, and M. Spranger, "Logic Tensor Networks," *Artificial Intelligence*, vol. 303, p. 103649, 2022.

[18] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," Journal of Computational Physics, vol. 378, pp. 686–707, 2019.

[19] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422-440, 2021.

[20] G. S. Misyris, A. Venzke, and S. Chatzivasileiadis, "Physics-informed neural networks for power systems," in *2020 IEEE Power & Energy Society General Meeting (PESGM)*, 2020.

[21] F. Fioretto, T. W. K. Mak, and P. Van Hentenryck, "Predicting AC optimal power flows: Combining deep learning and Lagrangian dual methods," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 01, pp. 630-637, 2020.

[22] M. Chatzos, T. W. K. Mak, and P. Van Hentenryck, "Spatial network decomposition for fast and scalable AC-OPF learning," *IEEE Transactions on Power Systems*, vol. 37, no. 1, pp. 653-662, 2022.

[23] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055-A3081, 2021.

[24] A. S. Zamzam and N. D. Sidiropoulos, "Physics-Aware Neural Networks for Distribution System State Estimation," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4347-4356, 2020.

[25] A. S. Zamzam and P. Baker, "Learning optimal solutions for extremely fast AC optimal power flow," in *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, 2020.

[26] F. Li and Y. Du, *Deep Learning for Power System Applications: Case Studies Linking Artificial Intelligence and Power Systems*, Springer, 2023.

[27] T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, and H. Zareipour, "Energy Forecasting: A Review and Outlook," *IEEE Open Access Journal of Power and Energy*, vol. 7, pp. 326-339, 2020.

[28] J. R. Vázquez-Canteli and Z. Nagy, "Reinforcement learning for demand response: A review of algorithms and modeling techniques," Applied Energy, vol. 235, pp. 1072–1089, 2019.

[29] Y. Wang, Q. Chen, T. Hong, and C. Kang, "Review of smart meter data analytics: Applications, methodologies, and challenges," *IEEE Transactions on Smart Grid*, vol. 10, no. 3, pp. 3125-3148, 2019.

[30] H. Wang, G. Li, G. Wang, J. Peng, H. Jiang, and Y. Liu, "Deep learning based ensemble approach for probabilistic wind power forecasting," *Applied Energy*, vol. 188, pp. 56-70, 2017.

[31] P. L. Donti and J. Z. Kolter, "Machine Learning for Sustainable Energy Systems," *Annual Review of Environment and Resources*, vol. 46, pp. 719-747, 2021.

[32] S. Karagiannopoulos, P. Aristidou, and G. Hug, "Data-driven local control design for active distribution grids using off-line optimal power flow and machine learning techniques," *IEEE Transactions on Smart Grid*, vol. 10, no. 6, pp. 6461-6471, 2019.

[33] A. S. Dobbe, O. Sondermeijer, D. Fridovich-Keil, et al., "Toward distributed energy services: Decentralizing optimal power flow with machine learning," *IEEE Transactions on Smart Grid*, vol. 11, no. 2, pp. 1296-1306, 2020.

[34] M. Saffari and M. Khodayar, "Spatiotemporal Deep Learning for Power System Applications: A Survey," *IEEE Access*, vol. 12, pp. 93623-93657, 2024.

[35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[36] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[37] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.

[38] X. Zheng, N. Xu, L. Trinh, D. Wu, T. Huang, S. Sivaranjani, Y. Liu, and L. Xie, "PSML: A multi-scale time-series dataset for machine learning in decarbonized energy grids," arXiv preprint arXiv:2110.06324, 2021.